Medical MythTriage: Can a Fast Decision Model Be Trusted to Pre-Screen Medical Claims?

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ABSTRACT

Health claims spread on social platforms far faster than human fact-checkers can review them, and careful verification against the medical literature does not scale. Platforms therefore face pressure to let a fast model pre-screen claims. We study whether such delegation can be made safe. We introduce Medical MythTriage, a two-tier system in which a fast decision model (the glancer) sees each claim once, with no search or tools, and settles it or abstains; a confidence gate sends abstentions to a slow verifier, which checks the claim against retrieved literature and returns a verdict with validated citations. We built two evaluation sets of 100 claims each, in Chinese and US English, with three source-verified citations per claim, and pre-registered the headline metric (coverage at 95% precision) and three failure criteria. Evaluating the glancer (gpt-4o-mini) on two external benchmarks, we find a sharp contrast. On HealthFC (327 expert-written claims), the 0.90 gate settled 15.9% of claims at precision 1.000; coverage at 95% precision was 0.165. On MEDFACT (92 fluent, LLM-generated claims), no threshold reached 95% precision (coverage 0.000; gate precision 0.712), meeting our pre-registered failure condition. The glancer's confidence tracks the fluency of a claim rather than its truth: delegation safety is a property of the claim distribution, not of the gate alone.

1 RESEARCH QUESTIONS OR PROBLEM SPECIFICATION

Social platforms face far more health claims than human fact-checkers can review, and careful per-claim literature checking does not scale — so in practice, fast automated pre-screening decides what most users see. This project asks: **can a small, fast model that sees a medical claim exactly once, with no search and no tools (a *glancer*: a fast decision model), be trusted to settle a useful share of claims on its own — provided it is at least 95% precise on the claims it settles, and abstains on everything else, sending those to slow, careful verification?** And how large is that share, for two claim pools where the answer matters: everyday Chinese health claims and US-English claims, including ones trending this week? The intended users are platform trust-and-safety pipelines and public-facing myth-busting formats in the style of Alipay's Zhizhenjia true-or-false feature, where a wrong fast answer about a vaccine rumor is far costlier than a slow one. This project began, in the approved proposal, as the same evaluation architecture applied to software-engineering agent decisions, graded by execution replay; at midway the domain moved to medical claims, while the research object — the delegation architecture, the precision gate, and the pre-registered failure criteria — is unchanged. What changed, and what Section 4 states plainly, is the kind of ground truth: literature consensus instead of deterministic replay.

2 ENVIRONMENT & DATA

Environment. The system operates on short health claims as they circulate on social feeds: one claim, shown once, with no surrounding thread. The user-facing environment is a quiz-format interface modeled on Zhizhenjia, Alipay's true-or-false feature: the user first commits to a judgment, then the evidence verdict is revealed with its citations, so disagreement between the fast judgment and the evidence is visible rather than asserted. The slow side of the environment is the published medical literature, accessed through PubMed.

Datasets. Two parallel evaluation sets of 100 claims were constructed for this project, one per language community (Table 1). The Chinese edition covers vaccines; infectious disease; cancer; cardiovascular; food & nutrition; supplements; sleep; exercise & weight; mental health;

and everyday beliefs (10 claims each). The US edition covers Thanksgiving; Christmas & holidays; everyday classics; food & diet; vaccines & flu season; supplements & wellness trends; winter health; US elections & civic facts; and items trending on X and on Reddit (10 claims each).

	Chinese edition	**US edition**
Claims	100 (10 categories × 10)	100 (10 categories × 10)
Verdict mix	68 myth / 22 supported / 10 partly true	60 Myth / 25 True / 15 Mixed
Citations	3 per claim, 300 total, each verified against PubMed (PMID recorded)	3 per claim, 300 total, each verified against its source

Table 1: The two project evaluation sets. Of the Chinese edition's 300 citations, 283 come from a top-journal whitelist (NEJM, The Lancet, JAMA, BMJ, Nature, Annals, Circulation, Cell, Cochrane) and 17 from top specialty journals where no whitelist venue covers the claim — flagged in the data. US trending items rest on the best evidence available at collection time and are reported as their own slice.

Three construction choices are deliberate. First, the supported/True slice (22–25%) is load-bearing: it is what makes a trivial "always say myth" baseline lose. Second, every claim carries exactly 3 citations, each checked to exist and to point in the direction the verdict requires — a rumor dataset with fabricated references would refute itself. Third, where the literature speaks in grades rather than verdicts (dose, population, timing), the claim is labeled partly true / Mixed rather than forced into a binary; results are reported both with and without the Mixed claims, since dropping exactly the hardest cases would inflate the headline number.

External validation sets. Two published benchmarks were added at midway, so that the first evaluation results do not rest on self-built data alone. (1) **HealthFC** (Vladika, Schneider, & Matthes, 2024): claims fact-checked by medical experts at Medizin Transparent (Cochrane Austria), in German and English. We evaluate on its 327 *decisive* claims (202 supported, 125 refuted), excluding its 423 not-enough-information claims, which carry no decisive ground truth (dataset CC BY-NC-ND 4.0, used unmodified). (2) **MEDFACT** (CCL 2026 evaluation task): Chinese claims extracted from LLM-generated medical answers, each labeled by whether retrieved evidence supports it. Its released training split is heavily skewed (of 5,064 claims: 3,386 supported, only 46 refuted), so we evaluate on a balanced 92-claim sample — all 46 refuted claims plus 46 supported claims sampled with seed 42 — mapping supported to true and refuted to myth, and setting aside the partially-supported, uncertain, and not-applicable labels, which describe an evidence relation rather than a verdict on the claim itself (dataset Apache 2.0). The two sets stress opposite surfaces: HealthFC claims are expert-phrased questions about often-obscure interventions, while MEDFACT claims are fluent, textbook-style statements — the register in which a false claim is hardest to spot by eye.

Success criteria and metrics. The headline metric is **coverage at 95% precision**: sweeping the glancer's confidence threshold, what is the largest share of claims it may settle alone while holding precision at or above 95%? Secondary metrics: Brier score and expected calibration error; cost per 1,000 correct settlements; and precision reported **per error direction**, because the costs are asymmetric — certifying a dangerous myth as true is far worse than flagging a true claim as a myth. All headline numbers will carry bootstrap confidence intervals (at n = 100 per set, a 95%-precision estimate can carry roughly ±8 points). Three failure criteria were pre-registered, so the result is informative even when the answer is "no": **K1′ (label trust)** — if two independent evidence-checkers, working from the citations alone, disagree on more than 20% of verdicts, the labels cannot be trusted and the project stops; **K2′ (task too easy)** — if the always-myth baseline or a keyword heuristic matches the glancers, the set is too easy or too leaked and must be rebuilt; **K3′ (delegation unsafe)** — if no glancer reaches 95% precision at any coverage, fast screening of medical claims is unsafe at any threshold, and that is the reported result.

3 Methods

Architecture. The system is a two-tier delegation pipeline. **Tier 1 — the glancer.** Each glancer sees each claim exactly once, with no retrieval and no tools, and returns a verdict plus a confi-

dence. The lineup carried over from the proposal: Jev (a commercial System 1 decision model), a self-hosted Nimble-9B pinned to its Apache-licensed v1 weights, and a prompted LLM baseline, alongside the trivial always-myth and keyword-heuristic baselines. The glancer evaluated in the midway experiments of Section 4 is the prompted-LLM glancer (gpt-4o-mini) — the same model that fronts the Tier 2 prototype; the Jev and Nimble-9B runs remain final-phase work. **The gate.** A confidence threshold routes each claim: above it, the glancer's verdict stands; below it, the claim is abstained on and escalated. Sweeping the threshold traces the coverage–precision curve from which the headline metric is read; this abstention design follows the selective-prediction framing (El-Yaniv & Wiener, 2010), and the two-tier check follows the process-verification idea that intermediate judgments deserve their own verification (Lightman et al., 2023). **Tier 2 — the slow verifier.** A retrieval-and-verification agent that checks an escalated claim against the literature and returns a verdict with citations. Tier 2 began in human form — it is the protocol by which the datasets were built — and an automated prototype now also exists: a glancer, gate, and verifier pipeline adapted from the Assignment 1 agent harness (a bounded search–read loop over the 535 documents cited by the datasets, with code-enforced search caps and citation-ID validation at exit, so a citation the model did not actually retrieve cannot survive into a verdict). It was smoke-tested live on November 6 (gpt-4o-mini via OpenRouter): a flu-shot claim was settled by the glancer at the 0.9 gate, and a novel claim found in neither dataset ("drinking warm salt water cures high blood pressure") escalated at glancer confidence 0.80 and was returned by the verifier as myth, citing three retrieved sources (NEJM 2001; BMJ 2009; NEJM 2021), with zero citations dropped at validation. These were demonstration runs; the first systematic evaluation is the external validation reported in Section 4. Live PubMed retrieval, beyond the datasets' own cited corpus, remains final-phase work. **The investigation's focus** is the gate: where the threshold must sit, for which glancer, in which language, before delegation is safe.

External validation protocol. Each external claim was shown to the glancer exactly once, unmodified. A *mixed* or *unknown* reply counts as an abstention, never as correct. For each set we report strict accuracy, accuracy among decisive (non-abstaining) answers, per-direction accuracy, the settle share and precision at the project's 0.90 gate, and coverage at 95% precision from the threshold sweep — the same protocol pre-registered for the project's own sets. The evaluation code and the full per-claim outputs are released with the project repository.

Prototype. Two working web apps, one per edition, implement the user-facing half of the pipeline: a quiz mode (commit to true/myth, then the verdict is revealed with its 3 citations and a plain-language explanation) and a browse mode (claims searchable by category and verdict). The prototype serves the 200 pre-verified claims from a table; it does not yet verify arbitrary new claims — the two datasets are its regression test for the final phase.

Perturbation twins (A1′). Each claim also appears as *wording twins* — the same claim (i) neutrally rephrased, (ii) translated (Chinese original vs. English translation), and (iii) rewritten as a sensationalized headline — with the verdict unchanged by construction and re-checked against the claim's citations. A glancer whose verdict flips under rewording is pattern-matching, not judging; the twin gap (flip rate) is reported as a result in its own right, and the translation twins double as the control for the language confound (Chinese claims, English-dominant models).

Label-agreement check (K1′ implementation). A second checker, given only the citations and not the original verdicts, independently re-verdicts every claim; agreement is reported as Cohen's κ alongside the raw disagreement rate that K1′ gates on.

4 Summary of Progress

Done at midway. (1) The scope decision above, taken November 5 with its rationale recorded. (2) Both datasets: 200 claims and 600 citations, every citation verified at the source before the claim entered the set. (3) Both prototype apps (quiz + browse), demoed in the accompanying video. (4) The evaluation design frozen: metrics, perturbation twins, label-agreement check, and pre-registered kill criteria K1′–K3′. (5) The Tier 2 verification-agent prototype described in Section 3, built and run live end-to-end on two claims (one glancer settlement, one escalated verification with validated citations). (6) The first systematic glancer evaluation, run November 6 on the two external validation sets (419 claims), reported next.

First results: external validation. Table 2 reports the headline numbers; the two sets behave oppositely. On HealthFC, the glancer abstained on 58% of claims and was correct on 92.0% of the claims it decided (98.1% of supported claims, 71.0% of refuted ones); at the 0.90 gate it made no errors at all. On the balanced MEDFACT sample, the same glancer, gate, and protocol identified only 18.8% of the refuted claims it decided on — 26 fluent false claims were certified as true, several at confidence 0.90 — and, as Table 2 shows, no threshold reaches 95% precision at any coverage: the pre-registered K3′ condition, met on the first external test. The contrast is the midway finding: **glancer confidence tracks how fluent and plausible a claim sounds, not whether it is true.** A threshold that is safe on obscure, expert-style claims is unsafe on fluent, LLM-style claims, so delegation safety is a property of the claim distribution, not of the gate alone, and results must be reported per dataset, never pooled. This is consistent with the construct-validity critique of forced verdicts in medical fact-checking (Joseph et al., 2026), and it converts the final phase's key question into a measurement: the *rescue rate* — how often Tier 2 evidence-checking overturns the glancer's confident errors.

	HealthFC	**MEDFACT (balanced)**
Claims evaluated	327 decisive	92 (46 supported / 46 refuted)
Abstention rate	58%	16%
Accuracy among decisive answers	92.0%	66.2%
Settle share at the 0.90 gate	15.9%	64.1%
Precision at the 0.90 gate	1.000	0.712
Coverage at 95% precision	0.165	0.000

Table 2: External validation of the prompted-LLM glancer (gpt-4o-mini), November 6, 2026. Per-direction accuracy: HealthFC 98.1% supported / 71.0% refuted; MEDFACT 100% supported / 18.8% refuted. Coverage at 95% precision is read from a threshold sweep, per the pre-registered protocol.

Not yet done, stated plainly. The glancer sweep on the project's own two sets (all glancers, plus wording twins) has not been executed; the midway numbers above are external validation, not results on the headline sets. The K1′ second-checker pass and its agreement statistic are likewise pending, as are the rescue-rate experiment, a human anchor (3–5 lay readers and one MCiM peer judging the same sets, to locate glancer performance against untrained and trained human glances), and live PubMed retrieval for Tier 2. Next steps, in order: run K1′; run all glancers on both project sets plus twins; measure the Tier 2 rescue rate on confidently-wrong claims; report coverage at 95% precision with the per-direction and with/without-Mixed breakdowns; then extend Tier 2 to live PubMed retrieval for unseen claims, with abstention as a first-class output.

Obstacles and concerns. The central one is the ground-truth change conceded above: execution replay is deterministic, while a literature verdict is a human consensus snapshot — experts can disagree, and consensus can drift. The mitigations are procedural, not rhetorical: a stated evidence hierarchy (meta-analyses and guidelines above single trials), the K1′ double-check with agreement reported, and per-direction precision so a soft label cannot hide inside a hard average. Second, calibration does not transfer across claim distributions — the external validation's central lesson — so a single global gate would silently import fluent-claim failures; per-distribution thresholds, or abstention by default on fluent registers, are required. Third, contamination: these are famous rumors, published fact-checks exist in training data, and a glancer may be recalling verdicts rather than judging claims; wording twins blunt this, and a held-out slice of novel, post-cutoff claims is the stronger answer. Fourth, size: at n = 100 per set the intervals are wide and calibration error is close to noise; both project sets are pilots, and the final-phase target is 300–500 claims. Fifth, the trending categories are by construction the least settled items in the set, and their verdicts may need revision as reporting settles. Finally, a framing caution: verdicts summarize published evidence and the product is not medical advice; and because a quiz repeats each myth before debunking it — which the misinformation literature warns can backfire through familiarity — the interface uses truth-first framing and never shows a myth without its verdict attached after answering.

5 References

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6 AI-Use Disclosure Statement

AI assistants were used in this project to help collect candidate claims and candidate sources, to build the prototype web applications, and to draft and edit this report and the evaluation scripts. Every citation in the datasets was checked against its source record (PubMed or the publisher) before the corresponding claim was admitted. All evaluation runs were executed by the author, and every number reported above is computed from those runs' logged per-claim outputs, released with the project repository. The research questions, evaluation design, verdict protocol, and all final decisions about content are the author's.